

ARAVIND R

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EDUCATION

B.Sc. in Biotechnology, Microbiology, Biochemistry Yuvaraja's College, University of Mysore	2011-2014
M.Sc. in Bioinformatics and Applied Biotechnology Institute of Bioinformatics and Applied Biotechnology (IBAB)	2014-2016
Internship at IBAB	2016-2018
PhD at the National Centre for Biological Sciences (NCBS)	2019-2024

RESEARCH EXPERIENCE

Supervisor: Dr S. Thiyagarajan – Intern (IBAB)

- Phenotype detection of missense mutations associated with mucopolysaccharidosis IV A using Machine Learning approaches.

Supervisor: Dr Ranabir Das – PhD (Computational Structural Biology) (NCBS)

- Plasticity of the proteasome-targeting signal Fat10 enhances substrate degradation.
- Open and closed state population bias observed in E3-independent and E3-dependent classes of E2 enzymes.
- Investigation of NEDD8 stability through long-chain hydrophobic amino acids.
- Structural rationale for BRCA2 mutation phenotypes in Malaysian Indian populations.

IT SKILLS

Programming Languages: Shell scripting & Python

Statistical packages: R

Operating Systems: Linux (Ubuntu, Fedora, Centos, Rocky) & Windows

COMPUTATIONAL SKILLS

- **Software used-** Amber, GROMACS, CHARMM, Autodock, Modeller, Schrodinger packages, UCSF Chimera, Pymol, RAST, automar & ccp4i and coot.
- **Protein structure/fold prediction-** Concepts and algorithms related to homology modelling, loop modelling, sequence-structure alignment (1D-3D), secondary structure prediction; structure refinement methods like energy minimisation.
- **MD simulation techniques-** Conventional MD simulations, Simulated Annealing, High-Temperature Simulations, SBM coarse-grained simulations, Umbrella Sampling, Accelerated Steered Molecular Dynamics (ASMD), Replica Exchange and coarse-grained self-organized-polymer model with side chains (SOP-SC) simulations.
- **Simulation analysis-** Principal Component Analysis (PCA), Free Energy Calculations, MMPBSA, Cytoscape, Structure Overlap Function & Residue Interaction Network.
- **Rational drug design-** Identifying active site/drug binding pockets. Protein-protein interactions. Docking algorithms and Virtual screening method in identifying potential lead molecules (Schrodinger Package).

FELLOWSHIPS

- Successfully cleared the **DBT-JRF** exam 2017 under Category-I.

PUBLICATIONS

- The Thermodynamic Properties of Fat10ylated Proteins Are Regulated by the Fat10ylation Site; Journal **ACS Omega**; PMID: [38799324](#)
- Plasticity of the proteasome-targeting signal Fat10 enhances substrate degradation; Journal **elife**; PMID: [38984715](#)
- Bacterial deamidases modulate ubiquitin structure and dynamics to dysregulate ubiquitin signalling. [Bio-archived](#)
- “Evolutionary Genomics of an Ancient Prophage of the Order Sphingomonadales” in Journal: **Genome Biology and Evolution** (GBE); PMID:[28201618](#).
- Epidemiological and ES cell-based functional evaluation of BRCA2 variants identified in families with breast cancer in Journal: **Human Mutation**; PMID: [33314489](#)

Under Preparation:

- Long-chain hydrophobic amino acids dictate stability in NEDD8.
- Conformational accessibility of K48 in Di-ubiquitin chains bound to UbE2k.
- Modelling and Dynamics of ORF61-E2D2 interaction Interface.

MINOR PROJECTS

- Characterization of Bacteriophage specific for Enterobacter Y1; Dr Shivakumar S, IBAB
- Identification and Characterization of Prophage elements in *Sphingomonas* Genome.
- Isolation and purification of L- asparaginase from *Ocimum tenuiflorum* and curry plants; Dr Bibha Chaudhary, IBAB.

PRESENTATIONS/ WORKSHOPS

- Presented a short talk titled- “Plasticity of the proteasome-targeting signal Fat10 enhances substrate degradation” at [ProUPS](#) conference
- Presented a poster on “Folding upon binding dictates active site structural dynamics in GP78” at the EMBO Structural Biology conference.
- Attended a hands-on training session on Human RNA-Sequencing: - Library Preparation, Sequencing- Illumina HiSeq 2500 and Data Analysis at IBAB.
- Presented a poster titled “Computational and structural studies on human iduronate-2-sulfatase (hIDS) missense mutants” at IBAB.
- Presented a Poster at IBAB on “Identification and Characterization of Prophage Elements in Sphingomonas Genome”.

DECLARATION

I hereby declare that the above-mentioned information is correct to my knowledge and I bear the responsibility for the correctness of the mentioned particulars.

Place: Bengaluru, India

Date: Feb 2025

Aravind R